

PHYSICS

This section contains multiple choice questions. Each question has four options, out of which exactly one is correct. Each correct answer carries +4 marks and each incorrect answer carries -1 mark.

1. A particle moves along a straight line with velocity $v = 3t^2 - 2t$. The displacement between $t = 0$ and $t = 2$ s is

- (A) 6 m
- (B) 4 m
- (C) 8 m
- (D) 2 m

2. Two resistors of 4 ohm and 6 ohm are connected in parallel across a 12 V supply. The total current drawn from the supply is

- (A) 2 A
- (B) 3 A
- (C) 5 A
- (D) 6 A

3. The dimensional formula of the coefficient of viscosity is

- (A) MLT^{-1}
- (B) $ML^{-1}T^{-1}$
- (C) $ML^{-1}T^{-2}$
- (D) MLT^{-2}

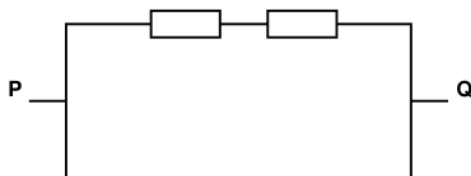
4. A body of mass 2 kg is raised through a height of 5 m. The work done against gravity, taking g as 10 m/s^2 , is

- (A) 50 J
- (B) 100 J
- (C) 150 J
- (D) 200 J

5. Light of wavelength 500 nm falls on a metal of work function 1.5 eV. The maximum kinetic energy of the emitted photoelectron is closest to

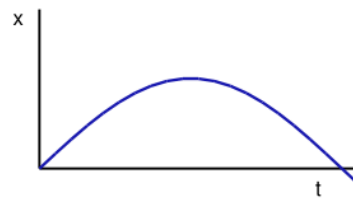
- (A) 0.98 eV
- (B) 1.24 eV
- (C) 0.48 eV
- (D) 2.48 eV

6. In the circuit shown in the figure, all resistors are identical and equal to R . The equivalent resistance between the terminals marked P and Q is



- (A) $R/2$
- (B) $2R/3$
- (C) $3R/2$
- (D) R

7. The graph shown depicts the variation of displacement with time for a particle. The velocity of the particle in the interval marked is



- (A) zero
- (B) constant and positive
- (C) constant and negative
- (D) increasing

The following questions require a numerical value as the answer. No options are provided. Round off the answer to the nearest integer where applicable.

8. A car accelerates uniformly from rest to 20 m/s in 8 s. Its acceleration in m/s^2 is

CHEMISTRY

This section contains multiple choice questions with exactly one correct option.

9. The number of sigma bonds present in a molecule of ethyne is

- (A) 2
- (B) 3
- (C) 4
- (D) 5

10. Which of the following has the highest first ionisation enthalpy?

- (A) Boron
- (B) Carbon
- (C) Nitrogen
- (D) Oxygen

11. The oxidation state of chromium in the dichromate ion is

- (A) +3
- (B) +5
- (C) +6
- (D) +7

12. A buffer solution is prepared by mixing equal volumes of 0.1 M acetic acid and 0.1 M sodium acetate. Given pK_a is 4.76, the pH of the solution is

- (A) 3.76
- (B) 4.76
- (C) 5.76
- (D) 7.00

The answer to the following question is a numerical value.

13. The pH of a 0.001 M solution of a strong monobasic acid is

MATHEMATICS

This section contains multiple choice questions with exactly one correct option.

14. If the roots of $x^2 - 5x + k = 0$ are real and distinct, then

- (A) $k > 25/4$
- (B) $k < 25/4$
- (C) $k = 25/4$
- (D) $k < 0$

15. The value of $\int_0^1 x e^x dx$ is

- (A) 1
- (B) $e - 1$
- (C) $2 - e$
- (D) e

16. The number of ways of arranging the letters of the word LEVEL is

- (A) 30
- (B) 60
- (C) 120
- (D) 20

17. If A is a 3×3 matrix with determinant 4, then the determinant of $2A$ is

- (A) 8
- (B) 16
- (C) 32
- (D) 64

Directions for Questions 18 to 19: Read the passage below and answer the questions that follow.

A thermodynamic system undergoes a cyclic process consisting of an isothermal expansion at temperature T_1 , an adiabatic expansion to temperature T_2 , an isothermal compression at T_2 , and an adiabatic compression returning it to its initial state. The working substance is one mole of an ideal monatomic gas, and all steps are carried out reversibly.

18. The efficiency of the cycle described above depends on

- (A) T_1 only
- (B) T_2 only
- (C) both T_1 and T_2
- (D) neither temperature

19. During the adiabatic expansion described in the passage, the quantity that remains constant is

- (A) temperature
- (B) pressure
- (C) entropy
- (D) volume

The answer to the following question is a numerical value.

20. The sum of the first 20 terms of the arithmetic progression 3, 7, 11, ... is